Engineering Chemistry

UNIT – 5 : Instrumental Methods & Applications

A friendly, exam-ready tour of spectroscopy and chromatography — from the EM spectrum and Beer's law to UV-Vis, IR, and HPLC, with clear block diagrams.

Topics in this unit

1. Electromagnetic spectrum
2. Beer–Lambert law
3. UV–Visible spectroscopy
4. IR spectroscopy
5. Chromatography & classification
6. HPLC: principle & instrumentation

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. Introduction & the Electromagnetic Spectrum

Modern chemistry rarely stops at test-tube colour changes. To find out *what* a substance is and *how much* is present, we shine energy on it and watch how it responds. These are the *instrumental methods of analysis*, and most of them are forms of *spectroscopy*.

DEFINITION

Spectroscopy: the study of the interaction of matter with electromagnetic (EM) radiation. By measuring the radiation a sample absorbs or emits, we learn about its structure and concentration.

EM radiation travels as a wave, described by its wavelength λ, frequency ν and energy E. These are tied together by one small but powerful equation:

$$E = h\nu = hc / \lambda$$

h = Planck's constant (6.626×10^{-34} J·s), c = speed of light. Note: shorter λ → higher ν → higher energy E.

When we arrange *all* EM radiations in order of increasing wavelength (decreasing energy), we get the *electromagnetic spectrum*.

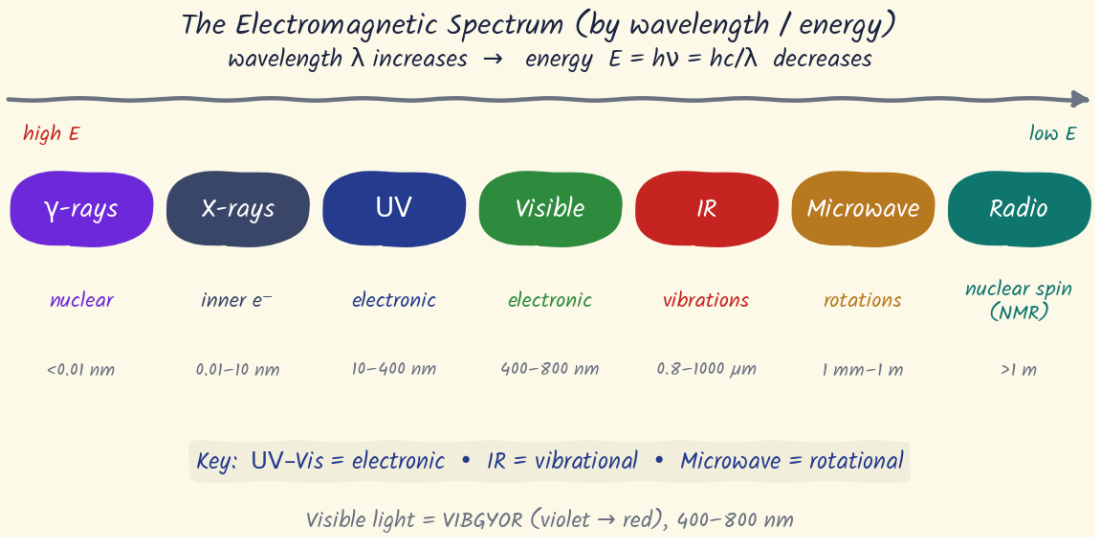


Fig 1 — The EM spectrum. Left → right: wavelength increases while energy decreases. Each region excites a different kind of change in matter.

- Order of regions: γ-rays → X-rays → UV → Visible → IR → Microwave → Radio.
- Moving right, wavelength increases and energy decreases ($E = hc/\lambda$).
- Visible light spans 400–800 nm (VIBGYOR: violet → red).
- The three regions we use most: UV-Visible → electronic transitions, IR → vibrations, Microwave → rotations.

> KEY POINT
> One idea unlocks the whole unit: each region of the spectrum carries a different amount of energy, so it excites a different motion in a molecule — electrons jump (UV-Vis), bonds vibrate (IR), molecules rotate (microwave).

2. Beer-Lambert Law

This is the single most important law for quantitative analysis — it tells us exactly how much light a solution soaks up. It combines two

simpler laws about a beam of monochromatic (single-wavelength) light passing through a solution.

DEFINITION

Lambert's law: the fraction of light absorbed is proportional to the path length (l) the light travels through the solution.

DEFINITION

Beer's law: the fraction of light absorbed is proportional to the concentration (c) of the absorbing solute.

Put together, they give the **Beer–Lambert law**:

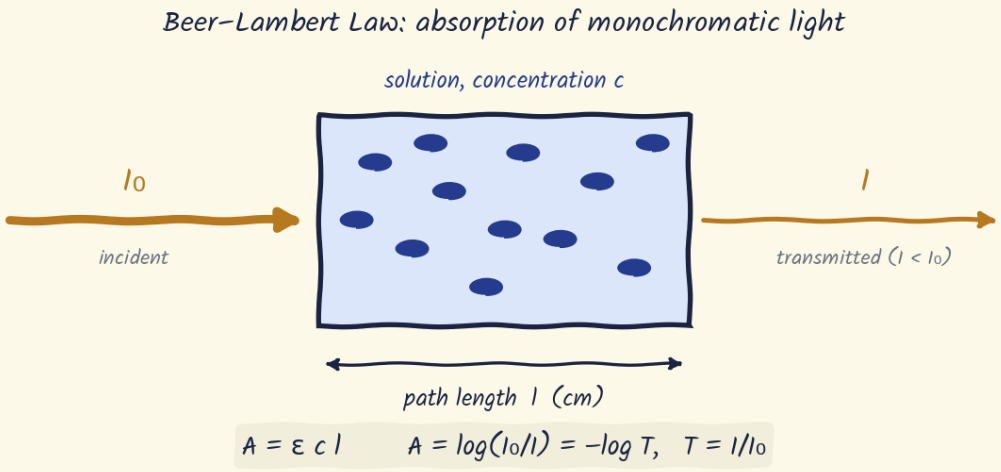


Fig 2 — Incident light I_0 enters a cuvette of path length l holding a solution of concentration c; weaker light I comes out.

$$A = \varepsilon c l$$

A = absorbance, ε = molar absorptivity (L mol^{-1} cm^{-1}), c = concentration (mol/L), l = path length (cm)

Absorbance and Transmittance

Two ways to describe the same measurement. **Transmittance (T)** is the fraction of light that gets through; **absorbance (A)** is how much is held

back.

$$T = I / I_0 \quad A = -\log T = \log (I_0 / I)$$

I_0 = incident intensity, I = transmitted intensity ($I < I_0$)

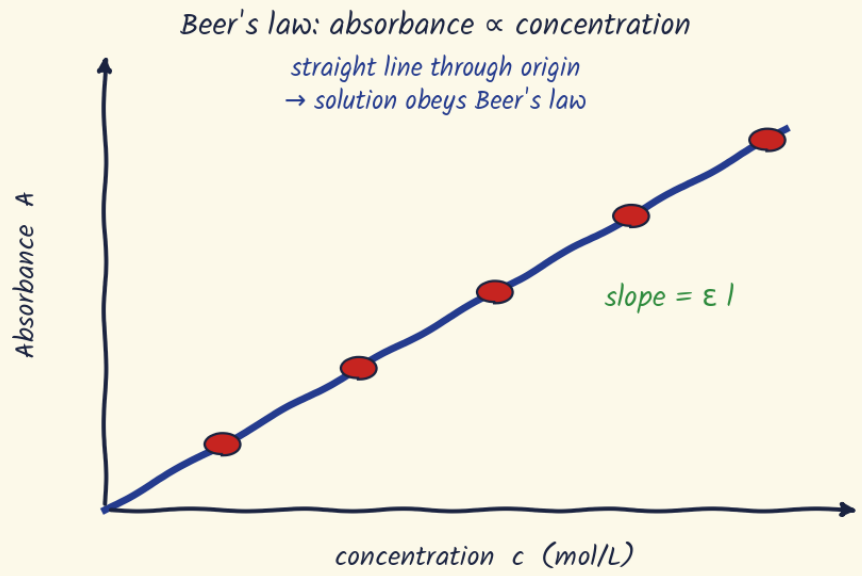


Fig 3 — Absorbance rises in a straight line with concentration (slope = εl). A line through the origin means the solution obeys Beer's law.

Limitations

- Holds only for dilute solutions (roughly $A < 1$); at high c the line bends.
- Needs truly monochromatic light.
- No chemical or physical change (association, dissociation, fluorescence) in the sample.

EXAMPLE

A 0.001 mol/L solution in a 1 cm cell has $\varepsilon = 2000$ L mol^{-1} cm^{-1}. Then $A = \varepsilon c l = 2000 \times 0.001 \times 1 = 2.0$ — doubling the concentration would double the absorbance.

3. UV-Visible Spectroscopy

Here we use UV light (200–400 nm) and visible light (400–800 nm). This radiation carries enough energy to push the molecule's valence electrons up to higher energy levels — so UV–Vis probes electronic transitions.

> DEFINITION
> **UV–Visible spectroscopy:** an absorption technique in which UV/visible radiation excites valence electrons from a lower to a higher molecular orbital. The wavelength absorbed tells us about the electronic structure.

Types of Electronic Transition

Electrons sit in σ, π and non-bonding (n) orbitals and can jump to the antibonding π* and σ* orbitals. The energy gap decides which wavelength is absorbed.

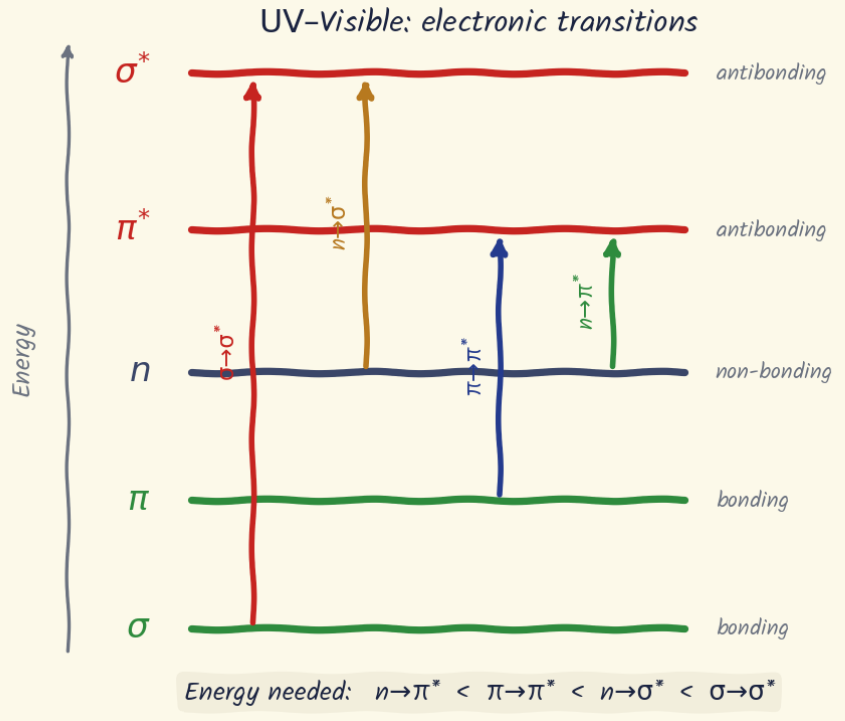


Fig 4 — The possible electronic transitions and their relative energies. n→π needs the least energy; σ→σ* needs the most.*

- $\sigma \rightarrow \sigma^*$ — highest energy (far UV); seen in saturated compounds like alkanes.
- $n \rightarrow \sigma^*$ — in molecules with lone pairs (e.g. alcohols, amines).
- $\pi \rightarrow \pi^*$ — in compounds with double bonds (alkenes, carbonyls).
- $n \rightarrow \pi^*$ — lowest energy; typical of C=O groups.
- Energy order: $n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$.

Chromophore & Auxochrome

DEFINITION

Chromophore: the group that actually absorbs UV-Vis light (the colour-giving unit). Examples: C=C, C=O, N=N and the benzene ring.

DEFINITION

Auxochrome: a saturated group with lone pairs (–OH, $-NH_2$, –Cl) that itself does not absorb, but when attached to a chromophore shifts and intensifies its absorption.

- Bathochromic (red) shift — absorption moves to longer λ.
- Hypsochromic (blue) shift — absorption moves to shorter λ.

Instrumentation

A UV-Visible spectrophotometer passes a single chosen wavelength through the sample and measures how much comes out, following the sequence below.

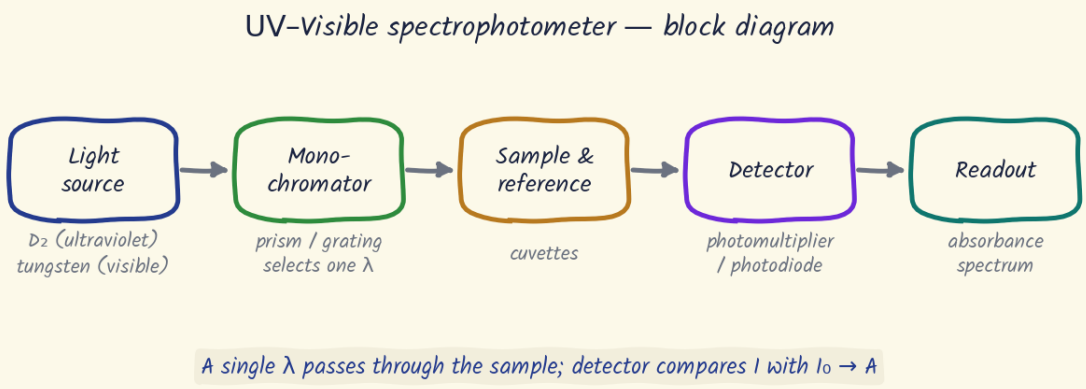


Fig 5 — Block diagram of a UV-Visible spectrophotometer.

- **Source** — deuterium (D_2) lamp for UV, tungsten lamp for visible.
- **Monochromator** — prism or grating that selects one wavelength λ.
- **Sample & reference** — matched cuvettes (sample vs blank).
- **Detector** — photomultiplier tube or photodiode converts light to current.
- **Readout** — records absorbance vs wavelength (the spectrum).

Applications

- **Quantitative estimation** of a substance using Beer's law ($A = \varepsilon c l$).
- **Detecting conjugation** — more conjugated double bonds absorb at longer λ.
- **Purity checking** and identification of unknown compounds.
- **Reaction kinetics** — following absorbance as a reaction proceeds.

> EXAMPLE
> $KMnO_4$ solution is purple because it absorbs green light (~525 nm). Its absorbance at that λ is directly proportional to concentration, so UV-Vis is used to measure how much $KMnO_4$ is present.

4. IR Spectroscopy

Infrared radiation carries less energy than visible light — not enough to move electrons, but just right to make bonds vibrate. So IR spectroscopy probes molecular vibrations and is the go-to tool for spotting functional groups.

DEFINITION

IR spectroscopy: an absorption technique using mid-IR radiation (4000–400 cm^{-1}, ≈ 2.5–25 μm) that excites the stretching and bending vibrations of covalent bonds.

KEY POINT

Selection rule: a vibration is IR-active only if it changes the dipole moment of the molecule. This is why symmetric molecules like O_2, N_2 and H_2 are IR-inactive — their stretch makes no dipole change.

Fundamental Modes of Vibration

Bonds move in two broad ways — the length can change (stretching) or the angle can change (bending / deformation).

IR vibrational modes of a 3-atom group (e.g. $-CH_2-$)

IR-active only if the vibration changes the dipole moment

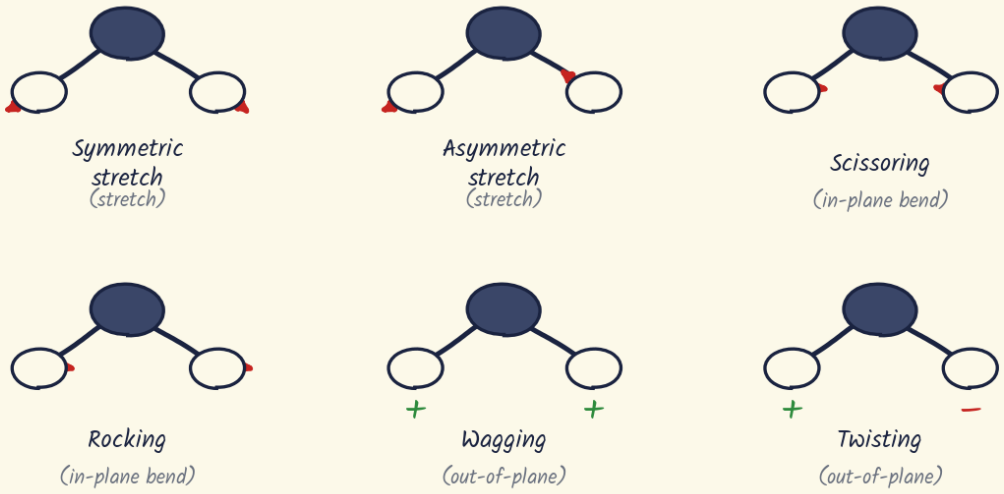


Fig 6 — Vibrational modes of a 3-atom group: two stretches and four bending modes (+ / − mean motion out of the plane).

- Stretching — symmetric and asymmetric (bond lengths change).
- Bending — scissoring, rocking (in-plane) and wagging, twisting (out-of-plane).

The number of fundamental vibrations depends on the number of atoms N:

Non-linear: $3N - 6$ Linear: $3N - 5$

e.g. H_2O (non-linear, N=3) → 3 modes; CO_2 (linear, N=3) → 4 modes

NOTE

The region below 1500 cm^{-1} is the fingerprint region — a complex pattern unique to each molecule, used to confirm identity (like a fingerprint).

Instrumentation

A dispersive IR spectrometer sends IR radiation through the sample and spreads it out to record how much is absorbed at each frequency.

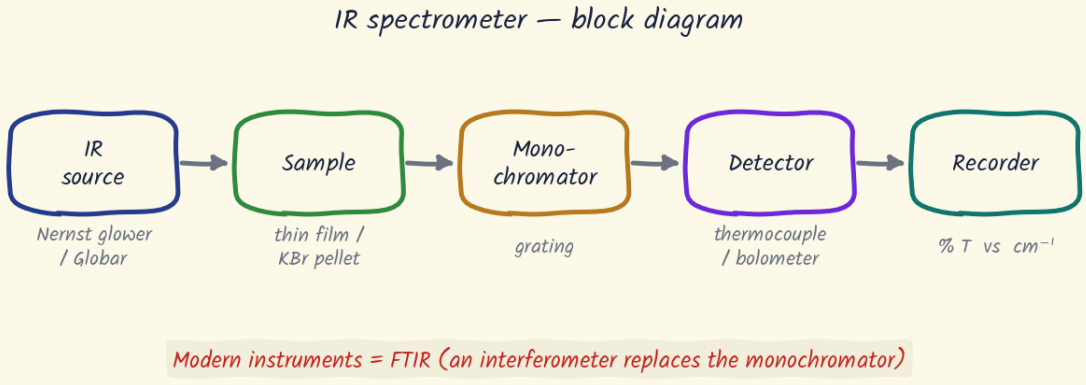


Fig 7 — Block diagram of an IR spectrometer.

- **Source** — Nernst glower or Globar (glows with IR when heated).
- **Sample** — as a thin film, liquid cell, or solid KBr pellet.
- **Monochromator** — a grating that disperses the IR beam.
- **Detector** — thermocouple or bolometer (senses heat).
- **Recorder** — plots % transmittance against wavenumber (cm^{-1}).

> NOTE
> Modern instruments are **FTIR** (Fourier-Transform IR): an **interferometer** replaces the monochromator, giving faster, more sensitive scans.

Applications

- **Functional-group identification** — e.g. a broad band near 3300 cm^{-1} = O–H; a sharp band near 1700 cm^{-1} = C=O.
- **Structure confirmation** of an unknown using the fingerprint region.
- Checking reaction progress and detecting impurities.

EXAMPLE

An alcohol and a ketone are easy to tell apart by IR: the alcohol shows a broad O–H stretch (~3300 cm^{-1}), while the ketone shows a strong C=O stretch (~1715 cm^{-1}).

5. Chromatography

The spectroscopic methods above identify and measure substances. Chromatography does something different — it separates a mixture into its pure components, so each can then be studied on its own.

DEFINITION

Chromatography: a technique that separates the components of a mixture based on their differential distribution between two phases — a fixed stationary phase and a moving mobile phase.

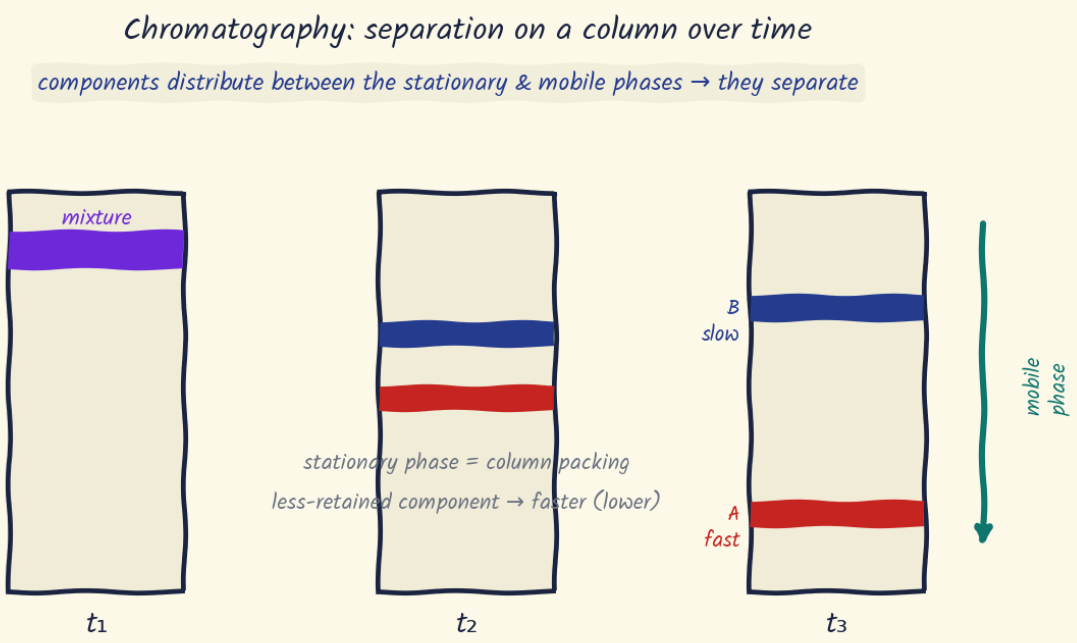


Fig 8 — As the mobile phase flows, each component spends a different time stuck to the stationary phase, so they travel at different speeds and separate.

- Stationary phase — stays fixed (paper, silica, a packed column).

- Mobile phase — flows and carries the sample (a liquid or gas).
- A component that clings more to the stationary phase moves slower; one that prefers the mobile phase moves faster → they separate.

$$R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$$

The retardation factor R_f is constant for a compound under fixed conditions.

Classification

Chromatography can be grouped two ways — by the mechanism of separation, or by the technique used.

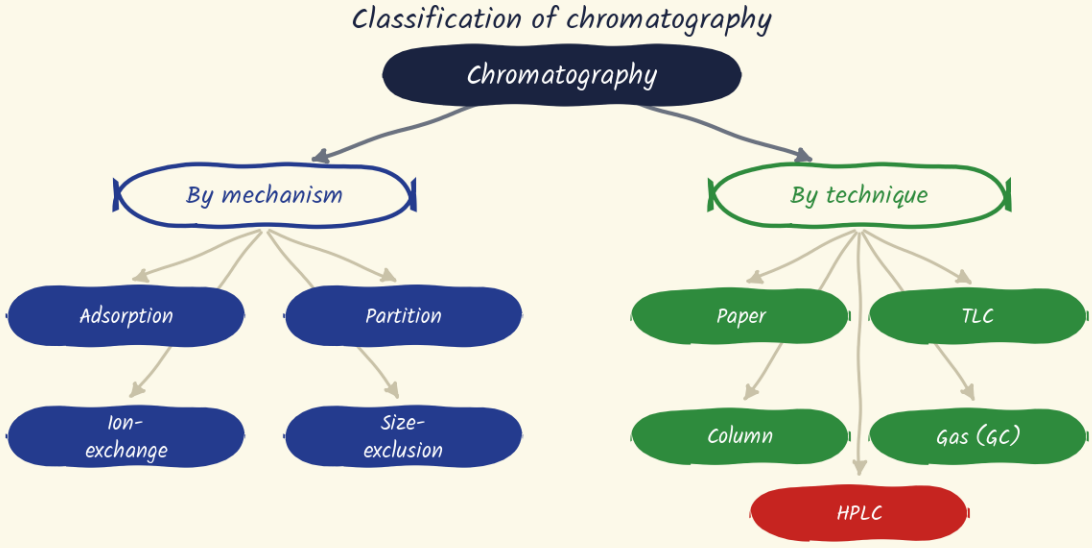


Fig 9 — Classification of chromatography by mechanism and by technique.

- By mechanism: adsorption, partition, ion-exchange, size-exclusion.
- By technique: paper, thin-layer (TLC), column, gas (GC) and HPLC.

EXAMPLE

In paper chromatography of ink, the paper is the stationary phase and the solvent is the mobile phase. The dyes separate into coloured spots, each with its own R_f value.

6. HPLC — High-Performance Liquid Chromatography

HPLC is column chromatography pushed to high performance. A liquid mobile phase is forced at high pressure through a column packed with very fine particles, giving fast, sharp and highly efficient separations.

DEFINITION

HPLC: a technique in which a liquid mobile phase is pumped under high pressure through a column of fine stationary-phase particles; the components separate by differential partition/adsorption and elute at different retention times, where a detector records them as peaks.

Instrumentation

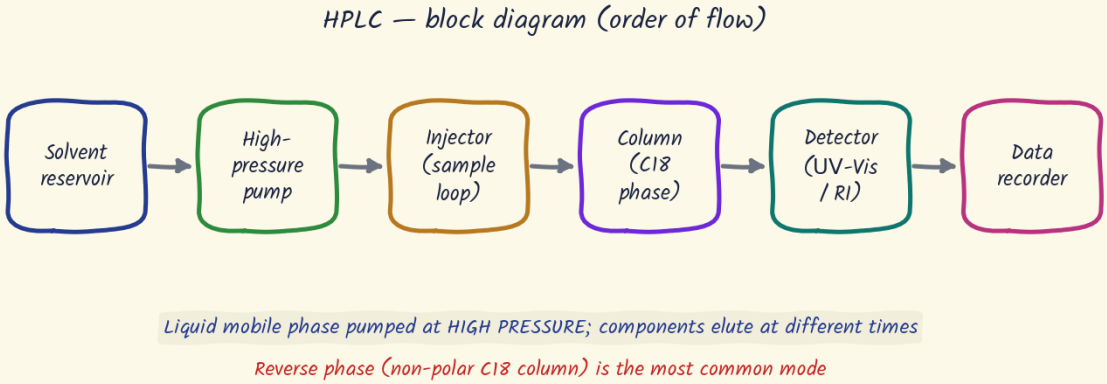


Fig 10 — Block diagram of an HPLC system, in the order the mobile phase flows.

1. Solvent reservoir — holds the mobile phase (eluent).
2. High-pressure pump — drives the solvent at constant high pressure.
3. Injector — introduces a fixed sample volume via a sample loop.
4. Column — packed with fine stationary-phase particles (the heart of HPLC).
5. Detector — UV-Vis, refractive-index (RI) or fluorescence, senses each component.

6. Data recorder — plots the chromatogram (peaks vs retention time).

Types of HPLC

Type	Stationary phase	Mobile phase	Note
Normal phase	polar (silica)	non-polar	polar compounds retained
Reverse phase	non-polar (C18)	polar	most common

Applications

- Separation and quantification of drugs (pharmaceutical analysis).
- Analysis of amino acids, proteins and biomolecules.
- Food and environmental testing (additives, pollutants).
- Purity testing and routine quality control (QC).

EXAMPLE

A pharma lab uses reverse-phase HPLC (C18 column) to check that a paracetamol tablet contains the correct amount of drug and no harmful impurities.

7. Applications & Importance

Instrumental methods are the backbone of modern analysis — fast, sensitive and needing only tiny samples:

- Pharmaceuticals: drug assay, purity and quality control by HPLC and UV–Vis.
- Structure determination: IR identifies functional groups; UV–Vis reveals conjugation.
- Environmental monitoring: detecting pollutants in water, air and soil.

- **Food & beverage:** checking additives, adulterants and nutrients.
- **Clinical & biochemical:** separating and measuring amino acids, proteins and metabolites.
- **Research & industry:** reaction monitoring, kinetics and material characterisation.

QUICK RECAP

- **Spectroscopy** studies how matter interacts with EM radiation; $E = h\nu = hc/\lambda$.
- EM spectrum order: $\gamma \rightarrow$ X-ray $\rightarrow$ UV $\rightarrow$ Visible $\rightarrow$ IR $\rightarrow$ Microwave $\rightarrow$ Radio (λ increases, E decreases).
- **Beer–Lambert law:** $A = \varepsilon cl$; $T = I/I_0$ and $A = -\log T$; valid for dilute solutions.
- **UV–Vis** excites valence electrons ($n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$); chromophores absorb, auxochromes shift absorption.
- **IR** excites vibrations (stretching & bending); IR-active only if the dipole moment changes; fingerprint region < 1500 cm^{-1}.
- **Chromatography** separates mixtures via a stationary + mobile phase; R_f = solute distance / solvent distance.
- **HPLC**: high-pressure liquid chromatography; reservoir $\rightarrow$ pump $\rightarrow$ injector $\rightarrow$ column $\rightarrow$ detector $\rightarrow$ recorder; reverse phase (C18) is most common.

EXAM-ORIENTED QUESTIONS

Q1. What is spectroscopy? Write the relation between energy, frequency and wavelength. [3 marks]

Ans. Spectroscopy is the study of the interaction of matter with electromagnetic radiation. Energy and radiation are related by $E = h\nu = hc/\lambda$, so shorter wavelength means higher energy.

Q2. Arrange the regions of the electromagnetic spectrum and state the change each causes. [3 marks]

Ans. In order of increasing wavelength: γ-rays → X-rays → UV → Visible → IR → Microwave → Radio. UV-Visible causes electronic transitions, IR causes molecular vibrations and microwave causes molecular rotations.

Q3. State and explain the Beer-Lambert law. Give its mathematical form. [5 marks]

Ans. Absorption of monochromatic light is proportional to both the path length (Lambert) and the concentration (Beer). Combined: $A = \varepsilon c l$, where A = absorbance, ε = molar absorptivity, c = concentration and l = path length.

Q4. Define absorbance and transmittance and relate them. [3 marks]

Ans. Transmittance $T = I/I_0$ is the fraction of light passing through. Absorbance A measures light held back: $A = -\log T = \log(I_0/I)$.

Q5. List the limitations of the Beer-Lambert law. [3 marks]

Ans. It holds only for dilute solutions, requires monochromatic light, and fails if the sample undergoes chemical or physical change (association, dissociation, fluorescence).

Q6. What are electronic transitions in UV-Visible spectroscopy? Give their energy order. [5 marks]

Ans. UV-Vis radiation excites valence electrons from σ, π or non-bonding (n) orbitals to antibonding π^* or σ^* orbitals. Energy order: $n\to\pi^* < \pi\to\pi^* < n\to\sigma^* < \sigma\to\sigma^*$.

Q7. Define chromophore and auxochrome with examples. [3 marks]

Ans. A chromophore is the group that absorbs UV-Vis light (C=C, C=O, N=N, benzene). An auxochrome (-OH, $-NH_2$, -Cl) does not absorb itself but shifts and intensifies the chromophore's absorption.

Q8. Draw/label the block diagram of a UV-Visible spectrophotometer. [5 marks]

Ans. Source (D_2 lamp for UV, tungsten for visible) → monochromator (prism/grating) → sample & reference cuvettes → detector (photomultiplier/photodiode) → readout.

Q9. When is a molecule IR-active? Why are O_2 and N_2 IR-inactive? [3 marks]

Ans. A vibration is IR-active only if it changes the dipole moment of the molecule. Homonuclear molecules like O_2 and N_2 have no dipole change on stretching, so they are IR-inactive.

Q10. Name the fundamental modes of molecular vibration studied in IR. [3 marks]

Ans. Stretching — symmetric and asymmetric; and bending — scissoring and rocking (in-plane), wagging and twisting (out-of-plane). Modes = $3N-6$ (non-linear) or $3N-5$ (linear).

Q11. What is chromatography? State its principle and classify it. [5 marks]

Ans. Chromatography separates a mixture by the differential distribution of its components between a stationary and a mobile phase. By mechanism: adsorption, partition, ion-exchange, size-exclusion. By technique: paper, TLC, column, GC, HPLC.

Q12. Explain the principle and instrumentation of HPLC and give two applications. [5 marks]

Ans. A liquid mobile phase is pumped at high pressure through a column of fine particles; components separate by differential partition and elute at different retention times. Set-up: reservoir → pump → injector → column → detector → recorder. Applications: drug analysis and protein/amino-acid separation.

Q13. What is a chromophore and an auxochrome? [3 marks]

Ans. A chromophore is the group that actually absorbs UV-Visible light (e.g. C=C, C=O, benzene). An auxochrome (e.g. –OH, $-NH_2$, –Cl) is an attached group that shifts the absorption to longer wavelength and makes it more intense.

Q14. List the main applications of UV-Visible and IR spectroscopy. [3 marks]

Ans. UV-Visible: quantitative estimation by Beer's law, detecting conjugation, purity checking and reaction kinetics. IR: identifying functional groups and confirming molecular structure using the fingerprint region.